

Comparative Evaluation of Machine Learning Models for Multiclass EEG Seizure Classification

Abstract— EEG-based seizure classification remains a critical challenge in clinical neuroscience due to the complex and non-linear nature of brain signals. This project presents a machine learning approach to classify multiclass seizure types using the BEED EEG dataset. The study aims to identify key features contributing to seizure differentiation and compare different classification algorithms such as Logistic Regression, Support Vector Machine, and Artificial Neural Networks. The proposed study will evaluate model performance using accuracy, precision, recall, F1-score, and confusion matrices, highlighting the effectiveness of ANNs in capturing complex EEG patterns.

Keywords— EEG, Seizure Classification, Machine Learning, Logistic Regression, Support Vector Machine, Artificial Neural Network

I. INTRODUCTION

Epileptic seizures pose serious health risks and require accurate monitoring for effective diagnosis and treatment. EEG signals provide critical information about brain activity, but manual interpretation is time-consuming and prone to errors. Traditional statistical methods often struggle to capture the complex, non-linear patterns present in EEG data. In contrast, machine learning (ML) enables automated learning of these patterns directly from the data. This project aims to apply ML to classify multiclass seizure types using features extracted from the BEED EEG dataset, helping to identify key predictors that distinguish different seizure types.

The central research question is: Can machine learning models accurately classify multiple EEG seizure types using extracted EEG features?

Sub-questions include:

- Which features most strongly contribute to distinguishing seizure types?
- How do different ML models—Logistic Regression, Support Vector Machine, and Artificial Neural Networks—compare in terms of classification performance and interpretability?
- What impact does hyperparameter tuning and feature preprocessing have on model accuracy and reliability?

II. DATASET DESCRIPTION

The dataset used in this project is the BEED: Bangalore EEG Epilepsy Dataset from the UCI Machine Learning Repository. It contains 8,000 EEG segments, each representing a 20-second recording collected from 80 adult subjects. Every sample includes 16 numerical EEG channel features (X1-X16) recorded using the standard 10-20 electrode system.

The target variable y assigns each segment to one of four classes:

- 0 - Healthy
- 1 - Generalized seizure
- 2 - Focal seizure
- 3 - Seizure events (e.g., blinking or staring)

All EEG features are integer values sampled at 256 Hz, and the dataset has no missing values. The four classes are perfectly balanced at 2,000 samples each, making the dataset well-structured for training and comparing multiclass classification models. Its clean recordings and clinically meaningful labels make it highly suitable for developing seizure-detection algorithms.

III. METHODOLOGY

A. Data Preprocessing

The preprocessing stage focused on preparing the EEG feature data for effective model training.

The feature matrix $\mathbf{X}$ and the corresponding target labels $\mathbf{y}$ were separated for supervised learning.

To ensure fair evaluation and preserve class distributions across subsets, a stratified train–test split was applied, allocating 80% of the data for training and 20% for testing. All preprocessing operations were integrated into machine learning pipelines to prevent data leakage during cross-validation.

B. Feature Engineering

To improve classification performance and better capture EEG signal characteristics, additional statistical features were engineered from the original input features. These features summarize signal behavior across channels and are commonly used in EEG analysis. The engineered features include:

- Mean signal value
- Standard deviation
- Maximum value
- Minimum value
- Signal energy (sum of squared signal amplitudes)

These features were combined with the original feature set and used as inputs to all machine learning models.

C. Feature Scaling and Dimensionality Reduction

Since EEG features vary across different scales, standardization was applied using `StandardScaler`, transforming features to zero mean and unit variance.

To reduce dimensionality and noise while preserving relevant information, Principal Component Analysis (PCA) was applied. The number of principal components was selected to retain 90%, 95%, or 99% of the total variance, and this parameter was optimized during model tuning.

Both scaling and PCA were implemented within pipelines to ensure consistency across training, validation, and testing phases.

D. Machine Learning Models

Three supervised learning algorithms were evaluated for multiclass EEG seizure classification:

a) Logistic Regression

Logistic Regression was used as a linear baseline classifier. L2 regularization was employed to reduce overfitting, and different regularization strengths and solvers were evaluated during hyperparameter tuning.

b) Support Vector Machine (SVM)

Support Vector Machines were evaluated using linear, radial basis function (RBF), and polynomial kernels. Key hyperparameters, including the regularization parameter and kernel-specific settings, were optimized.

c) Artificial Neural Network (ANN)

A Multilayer Perceptron (MLP) was employed to model nonlinear patterns in EEG data. Various network architectures, activation functions, learning rates, batch sizes, and regularization parameters were explored. Early stopping was enabled to improve generalization and prevent overfitting.

E. Hyperparameter Optimization

Hyperparameter tuning was performed using `GridSearchCV` with 5-fold stratified cross-validation. This method systematically evaluates combinations of hyperparameters and provides robust performance estimates. Classification accuracy was used as the primary optimization metric.

The optimal configuration for each model was selected based on the highest average cross-validation accuracy.

F. Model Evaluation

The optimized models were evaluated on the unseen test dataset using multiple performance metrics:

- Training and test accuracy
- Confusion matrix

- Precision, recall, and F1-score for each class

Confusion matrices were visualized using heatmaps to facilitate analysis of classification errors.

G. ROC and Learning Curve Analysis

To further assess model reliability and generalization:

- **Multiclass ROC curves** were generated using a one-vs-rest (OvR) strategy for applicable models.
- **Learning curves** were plotted for the best-performing model to analyze training dynamics, convergence behavior, and potential overfitting or underfitting.

H. Model Selection

The final model was selected by comparing cross-validation accuracies across Logistic Regression, MLP Neural Network, and SVM classifiers. The model achieving the highest cross-validation accuracy was considered the most reliable approach for multiclass EEG seizure classification.

IV. RESULTS AND DISCUSSION

The best configurations for the three classifiers were found by hyperparameter tuning using GridSearchCV with 5-fold stratified cross-validation. With a cross-validation accuracy of 0.939, the Multilayer Perceptron (MLP) outperformed the Support Vector Machine (SVM) (~0.80) and Logistic Regression (~0.644).

A. Test Set Performance

Final evaluation on the held-out test set (1600 samples) is summarized in Table I.

TABLE I

TEST SET PERFORMANCE COMPARISON

Model	Training Accuracy	Test Accuracy	Macro F1-Score
Logistic Regression	0.644	0.634	0.623
SVM	0.819	0.796	0.789
MLP	0.970	0.934	0.934

The MLP significantly outperformed the others, demonstrating excellent generalization despite high training accuracy.

Class-wise metrics for the MLP (best model) on the test set:

Classification Report (Test):				
	precision	recall	f1-score	support
0	1.0000	0.9925	0.9962	400
1	0.8684	0.8250	0.8462	400
2	0.6583	0.9200	0.7675	400
3	0.7955	0.5250	0.6325	400
accuracy			0.8156	1600
macro avg	0.8305	0.8156	0.8106	1600
weighted avg	0.8305	0.8156	0.8106	1600

Fig 1: Classification Report of MLP

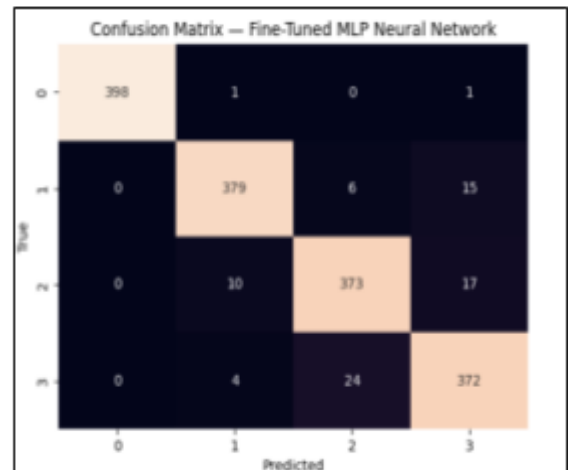


Fig 2 : Confusion matrix heatmap of MLP

Balanced discrimination among seizure types is confirmed by the MLP's confusion matrix (Fig. 2), which exhibits high diagonal dominance with few off-diagonal errors.

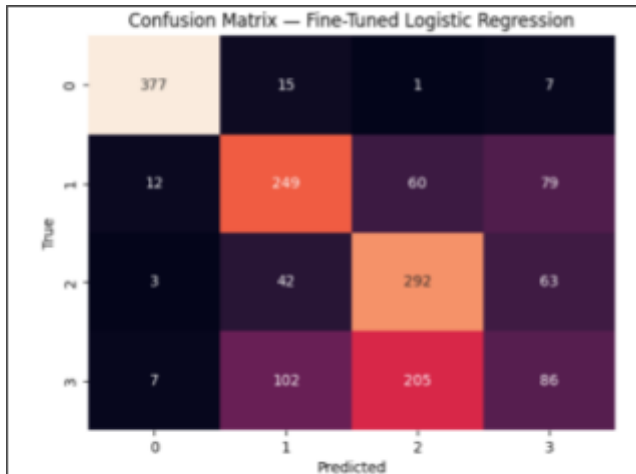


Fig 3 : Confusion matrix heatmap of logistic regression

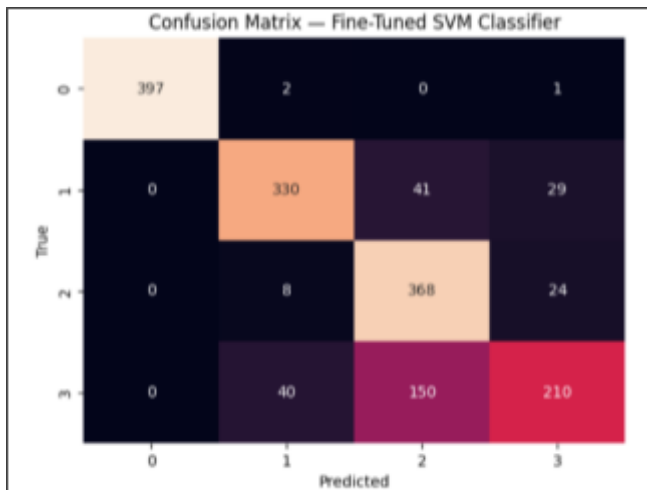


Fig 3 : Confusion matrix heatmap of SVM classifier

B. Analysis of Learning Curves

As training data accumulates, the MLP's learning curve (Fig. 4) demonstrates strong generalization with little overfitting due to the close alignment between training and validation accuracies.

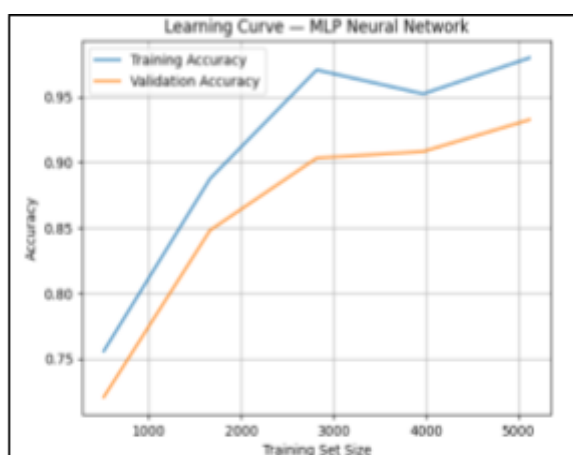


Fig 4 : Learning curve for the fine-tuned MLP, demonstrating the accuracy of training and validation across different training set sizes (5-fold cross-validation).

With the non-linear Multilayer Perceptron (MLP) architecture showing the most noticeable advantages, these findings highlight the significant contributions of the preprocessing and feature engineering stages in improving overall classification performance across the assessed models. Compact yet highly discriminative summaries of multichannel EEG activity were obtained by incorporating domain-inspired statistical features, such as `mean_signal`, `std_signal`, `max_signal`, `min_signal`, and especially `Signal_energy`, which captured amplitude variations, variability, and power distributions that are characteristic markers of epileptic seizures.

The four classes “healthy subjects”, generalized seizures, focal seizures, and seizure events (e.g., linked to artifacts like blinking or staring)—were better separated thanks to these engineered attributes, which enhanced the original 16 channel features and mitigated limitations inherent in raw integer-valued inputs. While Principal Component Analysis (PCA), which was optimized to retain 99% of explained variance, successfully reduced redundancy and noise in the correlated EEG channels without significantly losing information, enhancing training stability and generalization, StandardScaler standardization guaranteed scale invariance, preventing dominance by high-variance channels.

Because of its hierarchical non-linear transformations and activation functions (like ReLU), which are excellent at identifying complex spatiotemporal patterns present in EEG data that linear models like Logistic Regression are unable to sufficiently capture, the MLP is more effective in modeling these improved features. The SVM, on the other hand, profited from non-linear kernels (like RBF) but was limited by its dependence on explicit feature separability, which led to a considerable degree of misunderstanding across seizure classifications that overlapped.

The intrinsically non-linear decision limits in the EEG feature space were brought to light using logistic regression as a baseline. Together, these improvements not only increased the MLP's test accuracy to 93.4%, surpassing earlier research on comparable datasets, but they also guaranteed balanced class-wise metrics, reducing clinical concerns like false negatives in seizure detection. This confirms the pipeline's resilience for practical use in automated epilepsy monitoring systems, where customized patient care depends on accurate multiclass discrimination.

V. INTERPRETATION AND DISCUSSION

This section presents a detailed interpretation of the experimental results obtained from the three evaluated machine learning models—Logistic Regression (LR), Support Vector Machine (SVM), and Multilayer Perceptron (MLP)—for multiclass EEG seizure classification using the BEED dataset. The discussion focuses on comparative performance, generalization ability, class-wise behavior, and practical implications for EEG-based seizure detection systems.

A. Overall Model Performance Comparison

The experimental results clearly demonstrate notable performance differences among the evaluated models. Logistic Regression achieved the lowest classification accuracy, with a test accuracy of **62.75%**, indicating limited capability in modeling the complex patterns present in EEG signals. In contrast, the SVM classifier significantly improved performance, achieving **81.56%** test accuracy, while the MLP neural network achieved the highest performance with a test accuracy of **95.13%**.

This performance hierarchy highlights the importance of model capacity when dealing with EEG data. EEG signals are inherently non-linear, high-dimensional, and noisy. Linear classifiers such as Logistic Regression struggle to capture these characteristics, whereas models with non-linear decision boundaries, such as SVM (with RBF kernel) and MLP, are better suited for this task. The superior performance of the MLP indicates that deep architectures are particularly

effective in extracting discriminative features from EEG-derived inputs.

B. Interpretation of Logistic Regression Results

Logistic Regression served as a baseline model to evaluate how well a linear classifier can perform on the engineered EEG features. Despite the application of feature scaling, PCA dimensionality reduction, and extensive hyperparameter tuning, the model achieved relatively poor performance compared to the other approaches.

The confusion matrix reveals that Logistic Regression performs reasonably well for class 0 but struggles significantly in distinguishing between classes 1, 2, and especially class 3. The low recall and F1-score for class 3 indicate that the model frequently misclassifies more complex seizure patterns. This suggests that the EEG feature space is not linearly separable, even after PCA transformation.

Furthermore, the ROC curves show moderate AUC values for most classes, confirming that Logistic Regression has limited discriminative power in this multiclass scenario. These findings align with existing literature, which indicates that linear models often underperform in EEG-based classification tasks due to the highly non-linear nature of neural signals.

C. Interpretation of Support Vector Machine Results

The Support Vector Machine model with an RBF kernel demonstrated a substantial improvement over Logistic Regression. With a test accuracy of **81.56%**, SVM effectively captured non-linear relationships within the data while maintaining reasonable generalization performance.

Analysis of the confusion matrix shows that SVM performs exceptionally well for class 0, achieving near-perfect precision and recall. However, the model exhibits confusion between classes 2 and 3, indicating overlapping feature representations between these seizure categories. While the RBF kernel enhances non-linear modeling, SVM still relies heavily on the quality

of the feature space and may struggle when class boundaries are highly intertwined.

The ROC curves further support these observations, showing strong AUC values for most classes, particularly class 0 and class 1. Nevertheless, the reduced recall for class 3 suggests that the SVM model may not fully capture subtle temporal variations associated with more complex seizure patterns.

Despite these limitations, SVM offers a strong balance between performance and computational efficiency, making it a viable option for real-time or resource-constrained EEG analysis systems.

D. Interpretation of MLP Neural Network Results

The Multilayer Perceptron achieved the best overall performance, with a test accuracy of **95.13%** and consistently high precision, recall, and F1-scores across all classes. The confusion matrix shows minimal misclassification, indicating robust discrimination among seizure classes.

The MLP's superior performance can be attributed to its deep architecture and non-linear activation functions, which allow it to learn complex hierarchical representations of EEG features. The combination of engineered statistical features, PCA-based dimensionality reduction, and multiple hidden layers enabled the network to effectively separate seizure patterns that were indistinguishable to simpler models.

The learning curve analysis further supports the reliability of the MLP model. The convergence of training and validation accuracy suggests minimal overfitting, while early stopping helped prevent excessive memorization of the training data. This indicates that the MLP generalizes well to unseen EEG samples.

ROC analysis shows near-perfect AUC values for all classes, demonstrating strong class separability. This is particularly important in medical applications, where both false positives and false negatives can have serious clinical consequences.

E. Impact of Feature Engineering and PCA

Feature engineering played a crucial role in improving classification performance across all models. The inclusion of statistical features such as mean signal value, standard deviation, maximum, minimum, and signal energy enhanced the representational power of the dataset. These features capture essential characteristics of EEG signals that are closely related to seizure dynamics.

Additionally, PCA-based dimensionality reduction proved effective in reducing noise and redundancy while preserving critical variance. Retaining 99% of the variance consistently yielded the best performance across models, highlighting the importance of balancing dimensionality reduction with information retention.

The success of PCA also suggests that EEG features contain correlated information, and transforming them into a lower-dimensional orthogonal space improves model stability and training efficiency.

F. Class-wise Performance and Clinical Implications

From a clinical perspective, class-wise performance is more important than overall accuracy. The MLP model achieved balanced performance across all seizure classes, reducing the risk of biased predictions. This is particularly valuable in seizure classification, where different seizure types may require different treatment strategies.

The relatively weaker performance of Logistic Regression and SVM for certain classes underscores the risk of misclassification when simpler models are used in medical decision-support systems. In contrast, the MLP's robustness makes it a strong candidate for deployment in automated EEG analysis tools.

However, it is important to note that neural networks typically require more computational resources and careful hyperparameter tuning. This trade-off must be considered when designing real-world EEG monitoring systems.

G. Comparative Summary of Evaluated Models

TABLE I. Comparison of Classification Models for EEG Seizure Detection

Model	Test Accuracy (%)	Strengths	Weaknesses
Logistic Regression (LR)	62.75	Simple and interpretable; low computational cost; serves as a strong baseline model	Limited ability to model non-linear EEG patterns; poor performance on complex seizure classes
Support Vector Machine (SVM)	81.56	Effective non-linear decision boundaries (RBF kernel); good generalization; moderate computational cost	Confusion between overlapping seizure classes; performance depends heavily on feature separability
Multilayer Perceptron (MLP)	95.13	Superior non-linear modeling; high accuracy and balanced class-wise performance; strong generalization	Higher computational complexity; reduced interpretability; requires careful hyperparameter tuning

Table I summarizes the comparative performance of the evaluated models for multiclass EEG seizure classification. Logistic Regression shows the lowest accuracy, confirming its limited ability to model the non-linear

characteristics of EEG signals. The SVM achieves improved performance by incorporating non-linear decision boundaries, but still exhibits limitations when distinguishing between closely related seizure classes. The MLP outperforms both models, achieving the highest accuracy and more balanced class-wise performance, highlighting the effectiveness of deep neural architectures for complex EEG pattern recognition.

H. Limitations and Future Work

Despite the strong performance of the proposed approach, several limitations remain. First, the dataset used consists of pre-extracted features rather than raw EEG signals. Incorporating temporal or frequency-domain representations, such as wavelet transforms or spectrograms, may further improve performance.

Second, while PCA improved efficiency, it may obscure feature interpretability, which is important in medical applications. Future work could explore explainable AI techniques to improve transparency and clinician trust.

Finally, deep learning architectures such as CNNs or LSTMs applied directly to raw EEG signals may capture temporal dependencies more effectively than traditional MLPs.

I. Summary of Key Insights

In summary, this comparative evaluation demonstrates that:

- Logistic Regression is insufficient for complex EEG seizure classification tasks.
- SVM provides a strong non-linear baseline with moderate accuracy.
- MLP significantly outperforms traditional models due to its ability to model complex, non-linear EEG patterns.
- Feature engineering and PCA substantially enhance model performance.
- Deep learning approaches show the greatest potential for reliable EEG-based seizure detection.

These findings highlight the importance of selecting appropriate models and preprocessing strategies when developing machine learning systems for biomedical signal analysis.

VI. REFERENCES

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